

A GUT Microbiota Score Predicting for ACUTE GRAFT-Versus-Host Disease

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Although some studies have reported that intestinal microbiota is associated with aGVHD, they inadequately lacked a satisfactory method for precisely predicting aGVHD occurrence. We collected stool and blood samples at day 15 post-transplantation from 150 patients, who underwent allogeneic hematopoietic stem cell transplantation (allo-HSCT) from two independent centers. Stool microbiota was detected by 16S rRNA gene sequencing, inflammatory factors in blood were detected by multiplex immunoassays, and T lymphocyte subsets were examined by flow cytometry. A gut microbiota score (GMS) from a LASSO (least absolute shrinkage and selection operator) model using the gut microbiota was developed and validated to predict aGVHD. The discovery cohort included most of the identified bacterial families. As determined by area under the curve (AUC) analysis of ROC plots, the GMS could predict aGVHD in grades II-IV (AUC = 0.904, $P < 0.0001$). Furthermore, the GMS model in the validation set also demonstrated a high AUC that was consistent with the discovery set (AUC = 0.887, $P < 0.0001$). The results also indicated that the ratio of Treg/Th17 cells in the low GMS subgroup was higher compared with the high GMS subgroup ($P = 0.012$), and the validation set was consistent with the discovery set ($P = 0.003$). In addition, high cytokine levels (including IL-1 β , IL-6, IL-17, and TNF- α) were associated with high GMS. In conclusions, the GMS at neutrophil engraftment could predict the subsequent aGVHD, and

it is a potential and novel method. The GMS was associated with the inflammatory factor levels and Treg/Th17 balance.

Disclosures

No relevant conflicts of interest to declare.

Author notes

*Asterisk with author names denotes non-ASH members.

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